

Technical Note

Row Boundary Crossing Functionality in CellularRAM™ Memory

Introduction

Micron's CellularRAM™ devices are designed to be backward-compatible with 6T SRAM and early-generation asynchronous and page PSRAM, and are based on a fixed row size. Memory controllers must handle boundary crossings between consecutive rows.

This technical note describes the conditions that exist at the row boundary for the WAIT pin and burst READ/WRITE operations, focusing on row boundary crossing (RBC) as supported on CR 1.0-compliant devices. RBC support on other Micron PSRAM devices is discussed in Appendix A.

Row Boundary Crossing

RBC allows a memory controller to continuously burst across all locations within a CellularRAM device without concern for where the address is in relation to the boundary of the row. For the examples provided in this technical note, the device is configured as follows:

- Starting address = 0x00007Ch
- Latency code (LC) = 3
- Burst wrap = off
- Wait polarity = active LOW
- Burst length = continuous

Key sections of the bus configuration register (BCR):

- BCR[14] – Variable or fixed latency configuration
- BCR[13:11] – LC selection
- BCR[10] – WAIT polarity selection
- BCR[8] – WAIT configuration
- BCR[3] – Burst WRAP configuration
- BCR[2:0] – Burst length

Note: In the burst WRITE timing diagrams, data increments from the first cycle to help the reader understand when data is written as an RBC occurs. This would not occur in an actual system design.

WAIT Pin

The WAIT pin provides the following functionality:

- Provides data-valid feedback during burst READ and WRITE operations
- Arbitrates collisions between REFRESH and READ/WRITE operations
- Masks the delay associated with opening a new internal page
- Notifies the memory controller of an RBC

The WAIT pin plays an integral part when the device crosses the row boundary, but it does not always signify the point at which the data is written as the access crosses the boundary.

Four-Word Burst WRITE/READ, No Wrap, No RBC

When a burst WRITE commences (see Figure 1), there are a number of key areas of interest:

- The burst WRITE is identified and the first data on the DQ bus appears after the appropriate latency code. Assume that there is no refresh collision on this variable latency access.
- The BCR[8] setting will dictate when WAIT asserts and deasserts.
- On subsequent clock cycles, the device will increment the internal address counter and the CellularRAM device will provide the required data (see Table 1).
- The device is deselected before the internal burst counter increments to the next address (see “Point A” in Figures 1 and 2).
- The WAIT pin goes High-Z after CE# goes HIGH.
- Differences in the transition of the WAIT pin are only seen before the first data cycle. WAIT will go High-Z in relation to when CE# goes HIGH.

Figure 1: Four-Word Burst WRITE, No Wrap, No RBC

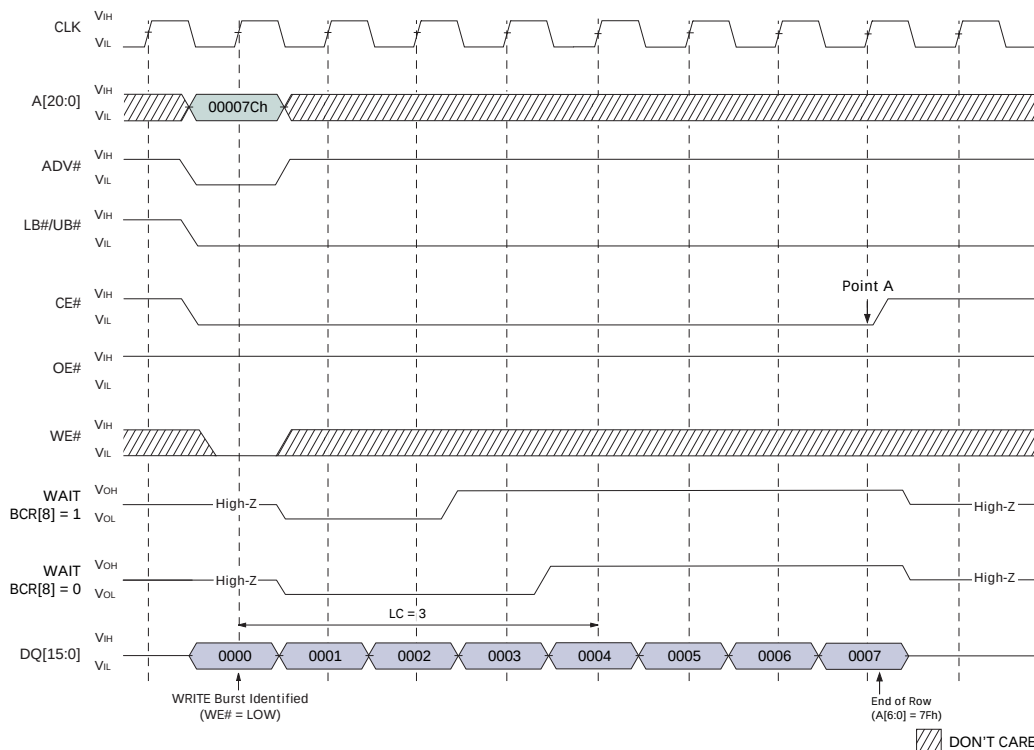
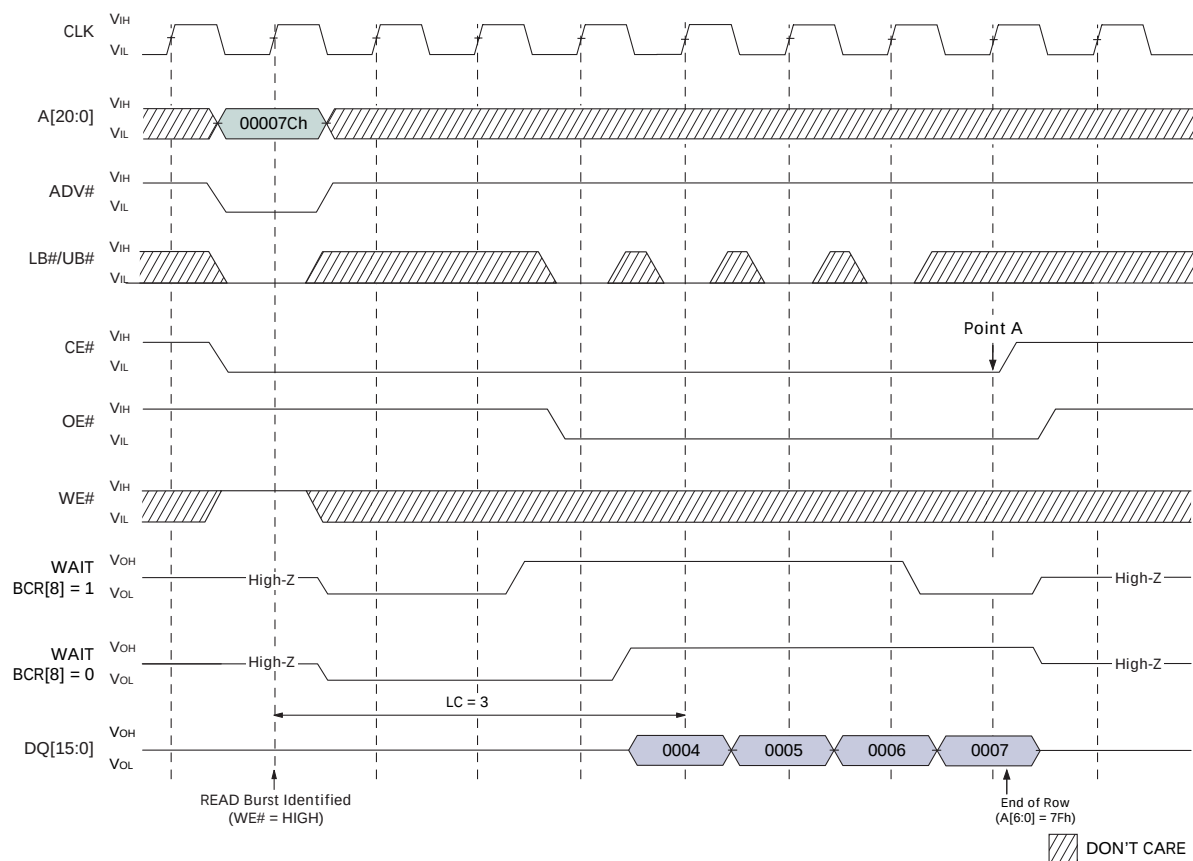


Table 1: Data vs. Address Locations (Prior to Point A)

Address	Write Data	Read Data
0x00007Ch	0x0004h	0x0004h
0x00007Dh	0x0005h	0x0005h
0x00007Eh	0x0006h	0x0006h
0x00007Fh	0x0007h	0x0007h

The result is that the data will be written to the memory locations. See Figure 2 on page 3 for a READ of the same locations.

Figure 2: Four-Word Burst READ, No Wrap, No RBC



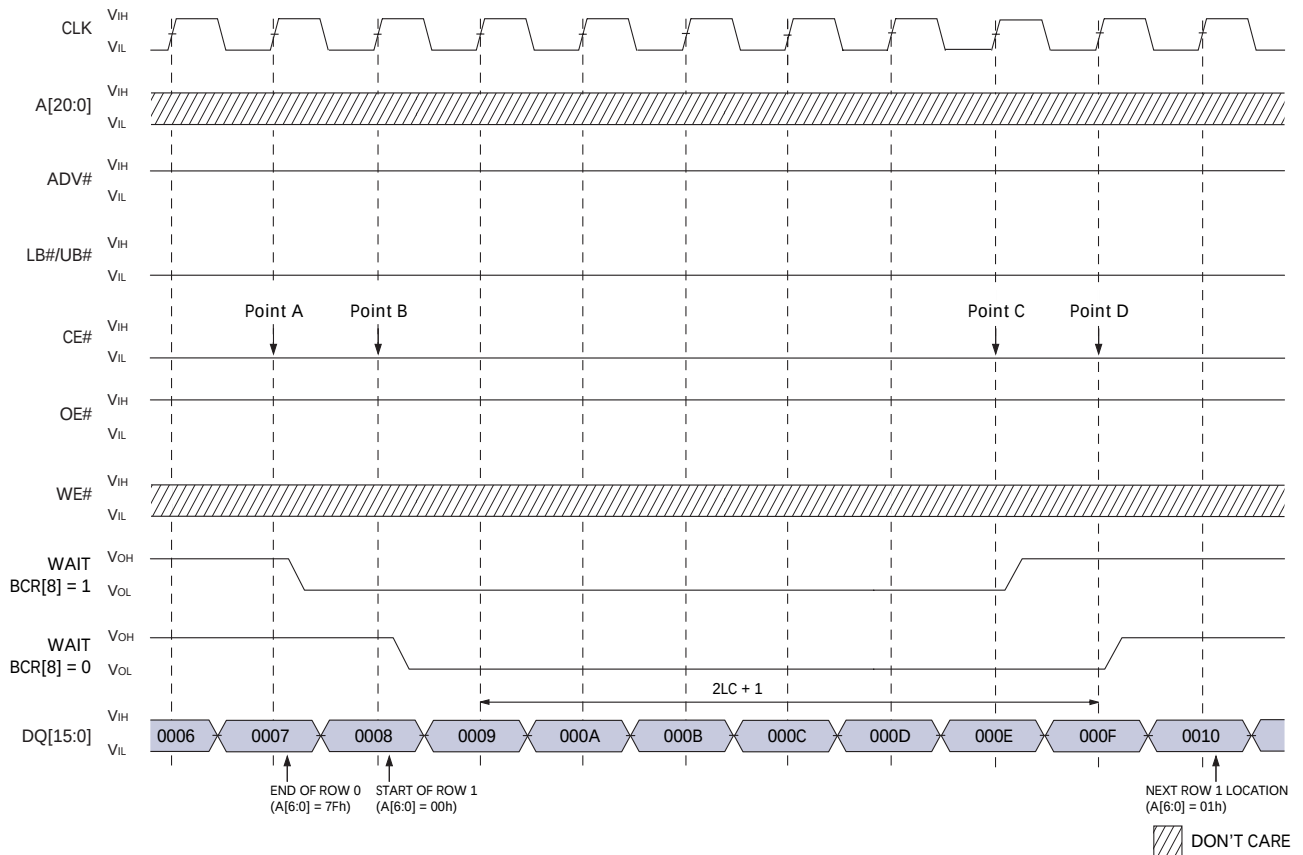
Four-Word Burst WRITE/READ, No Wrap, RBC Not Complete

In this section we examine what happens when the memory controller must cross from one row to the next during a burst access. For both burst read and burst write accesses, the controller must monitor WAIT to ensure that there is no loss of data.

RBC Burst WRITE

An RBC burst WRITE is shown in Figure 3. Whether the first location of the next row is written will depend on how long the device is selected ($CE\# = \text{LOW}$). This is defined as the point after the memory controller write to column 0, row 1 has occurred ("Point B").

Figure 3: Burst WRITE, RBC Occurs ($LC = 3$)



Notes: 1. Data increment order pictured is for concept illustration only and does not represent actual system design behavior; see page 1 note.

The options for burst WRITE are:

- If $CE\#$ remains asserted for less than $(2LC + 1)$ clock cycles after Point B, then the WRITE to column 0, row 1 will fail and the data will not be changed.
- If $CE\#$ remains asserted for $(2LC + 1)$ clock cycles after Point B, then the WRITE to column 0, row 1 will succeed and the data in column 0, row 1 will be changed.
- If $CE\#$ remains asserted for greater than $(2LC + 1)$ clock cycles after Point B, then the WRITE to column 0, row 1 will again succeed and the data in column 0, row 1 will be changed. Normal burst WRITE operation will commence and subsequent columns in row 1 will be written.

To better understand the results of deasserting CE# during a burst WRITE, see Tables 2 and 3, which show the correlation of values for each location in subsequent rows. Table 2 shows the values before and after a successful RBC, while Table 3 shows the results of a burst WRITE when CE# is deasserted at different points in the burst WRITE cycle.

Table 2: Burst Write Data vs. Address Locations

Memory Location	Initialized Data	Required Write Data
0x00007Ch	0x00FFh	0x0004h
0x00007Dh	0x00FFh	0x0005h
0x00007Eh	0x00FFh	0x0006h
0x00007Fh	0x00FFh	0x0007h
0x000080h	0x00FFh	0x0008h
0x000081h	0x00FFh	0x0010h

Table 3: RBC Burst Read Results

RBC Burst Write Conditions ¹	Burst Read Data		
	0x00007Fh	0x000080h	0x000081h
CE# LOW < 2LC + 1	0x0007h	0x00FFh	0x00FFh
CE# LOW = 2LC + 1	0x0007h	0x0008h	0x00FFh
CE# LOW > 2LC + 1	0x0007h	0x0008h	0x0010h

Notes: 1. LC defined by BCR[13:11].

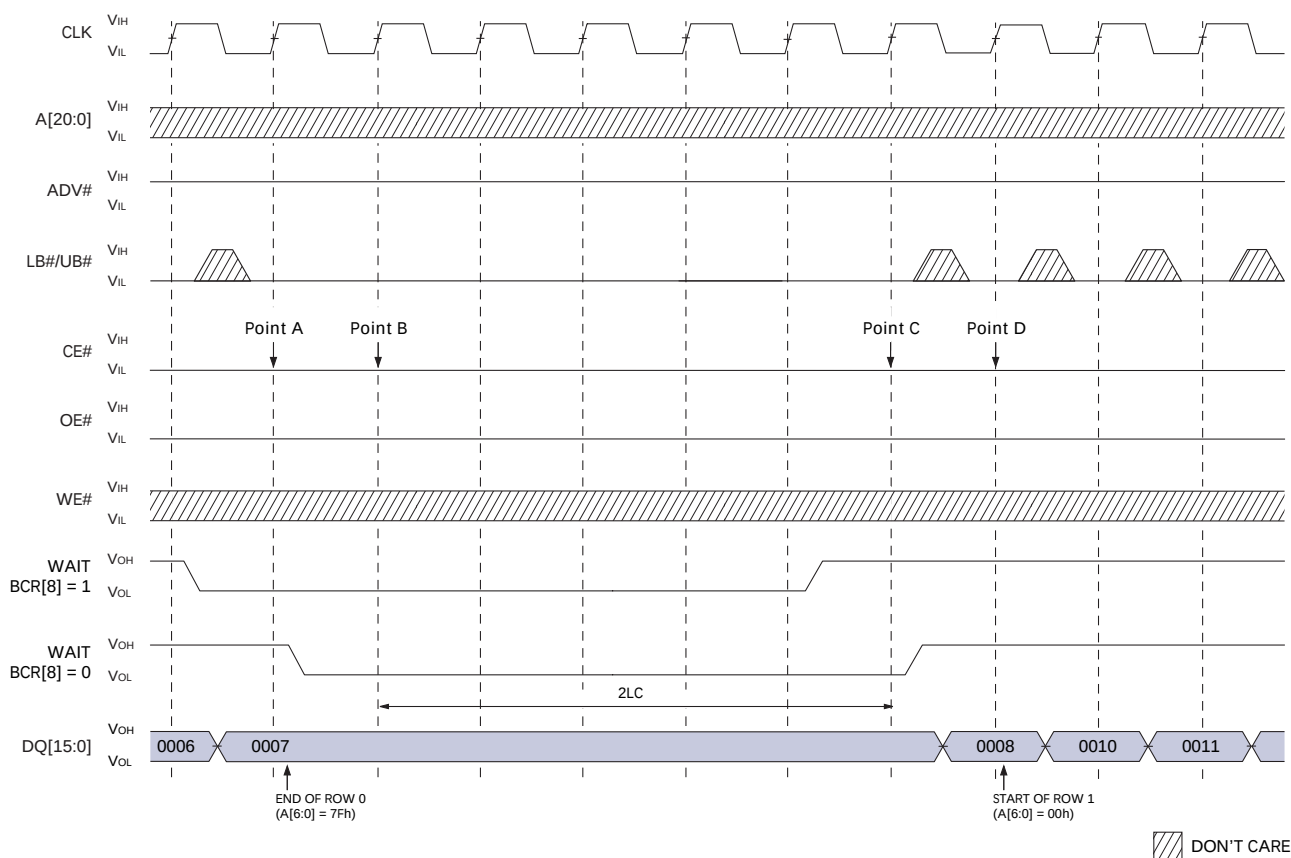
RBC Burst READ

A successful RBC burst READ that provides the data values per address shown in Table 2 on page 5 will look, from a signal point of view, like Figure 4.

The main points to consider in Figure 4 are:

- The WAIT pin will assert for 2LC during the RBC.
- If OE# is kept valid (LOW) during the RBC, then the value 0x0007h (address = 0x00007Fh) will remain on the data bus (DQ[15:0]).
- The burst READ of row 1 will continue sequentially after WAIT deasserts.

Figure 4: Burst READ, RBC Occurs (LC = 3)

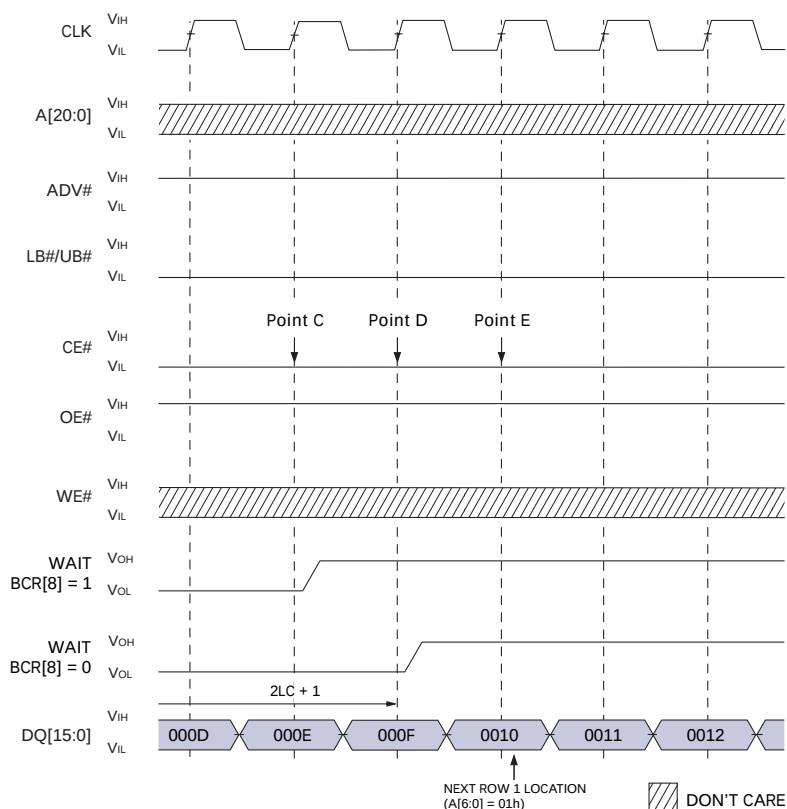


Four-Word Burst WRITE/READ, No Wrap, RBC Complete

Figure 5 shows the status of the burst WRITE once the RBC has completed. As shown previously, the value at column 0, row 1 will be written if CE# remains asserted for $(2LC + 1)$ clock cycles; this is “Point D.”

Once RBC is complete, the subsequent locations (columns) in row 1 will be changed by the burst WRITE command. The result of the subsequent burst READ is shown in Figure 6 on page 8, and the values are included in Table 4.

Figure 5: Four-Word Burst WRITE, No Wrap, RBC Complete

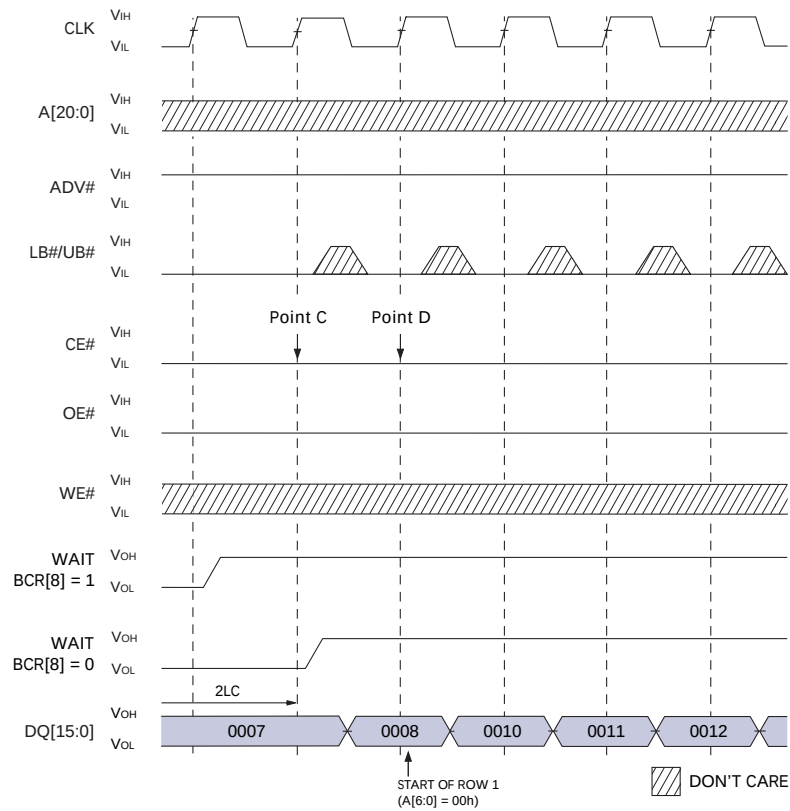


Notes: 1. Data increment order pictured is for concept illustration only and does not represent actual system design behavior; see page 1 note.

Table 4: Data vs. Address Locations, RBC Complete

Memory Location	Required Write Data
0x00007Ch	0x0004h
0x00007Dh	0x0005h
0x00007Eh	0x0006h
0x00007Fh	0x0007h
0x000080h	0x0008h
0x000081h	0x0010h
0x000082h	0x0011h
0x000083h	0x0012h

Figure 6: Four-Word Burst READ, No Wrap, RBC Complete



Other Burst Access Considerations

Burst WRITE Through Multiple Rows

If a controller needs to write to more than one row of data, the controller design may need to consider the following conditions:

- READ/WRITE One Location Burst (Last Column in a Row)
- Dealing with Losing (First Column of a Row) Data

READ/WRITE One Location Burst (Last Column in a Row)

The use of the CellularRAM device has evolved to include the possibility of a device driver requiring access to one location anywhere within the row. This means that the controller will need to handle the slight difference in the WAIT functionality if the location is the last column of the row.

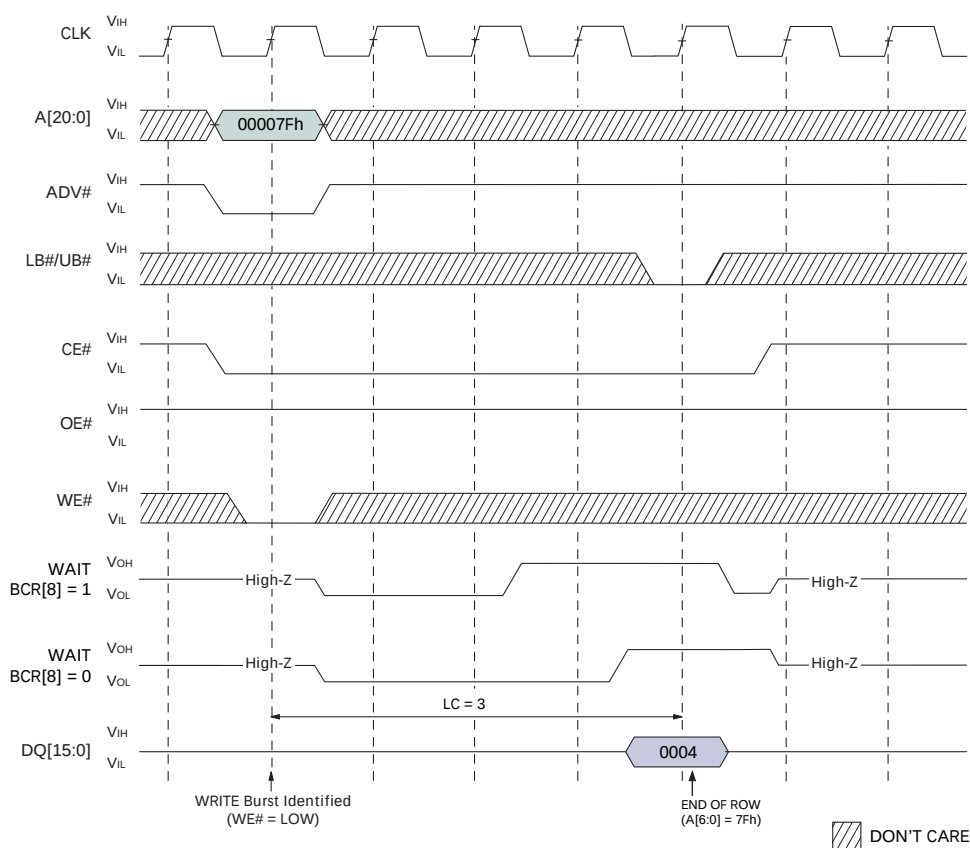
The following two sections—“Burst WRITE” and “Burst READ”—discuss the device characteristics when configured for burst no wrap ($\text{BCR}[3] = 1$). When a device is configured for burst wrap within the burst length ($\text{BCR}[3] = 1$), the WAIT pin will not assert when the last column of the current row is accessed; the next location to be accessed will be column x of the current row, where $x = 127 - \text{burst length}$.

Burst WRITE

For a burst write access from column 0 through column 126 of a row, the WAIT pin will function as shown in Figure 1 on page 2. The WAIT pin will assert at the start of the burst access and then deassert when device data is ready. WAIT will transition High-Z after CE# is taken inactive by the memory controller.

For a single burst write access to column 127, the WAIT pin will again assert at the start of the burst, as normal. The WAIT deassert will occur with the part “thinking” that it is doing a RBC. Therefore, the BCR[8] setting will dictate when the WAIT pin deasserts, with this change being referenced to the next clock edge after the access to column 127. Figure 7 shows this behavior.

Figure 7: Single Burst WRITE Starting at Column 127

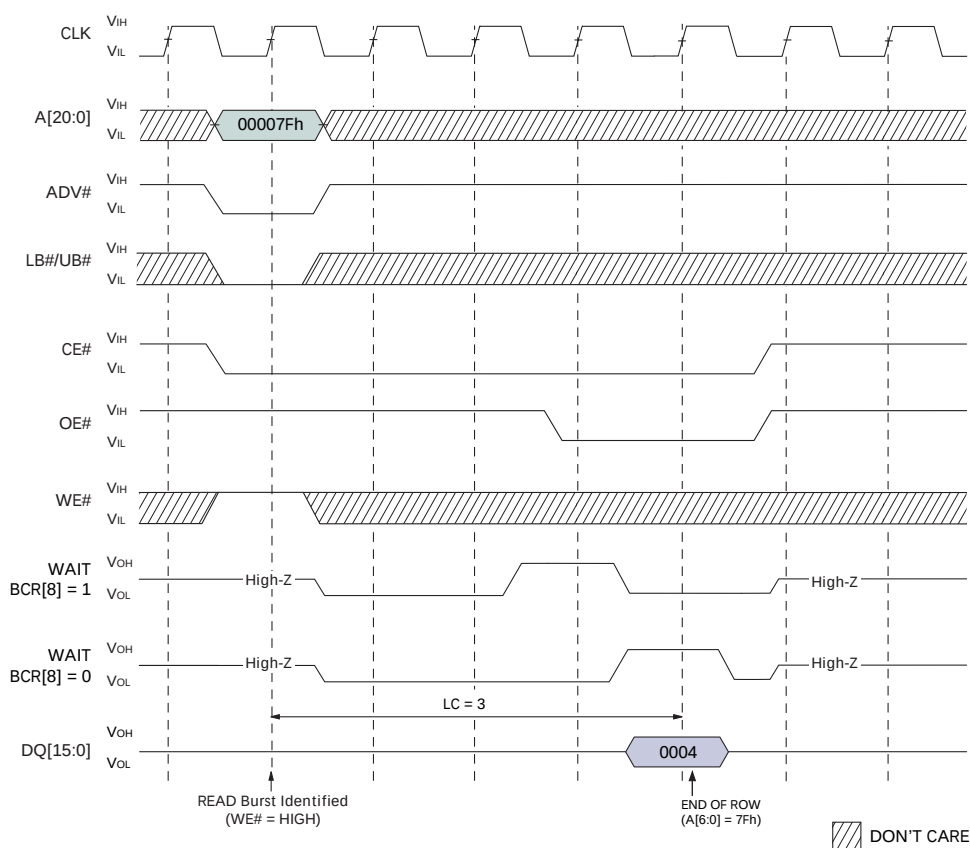


Burst READ

For a burst read access from column 0 through column 126 of a row, the WAIT pin will function as shown in Figure 2 on page 3. The WAIT pin will assert at the start of the burst access and then deassert when the device data is ready. WAIT will transition High-Z after CE# is driven inactive by the memory controller.

For a single burst read access to column 127, the WAIT pin will again assert at the start of the burst as normal. The WAIT assert will occur with the part “thinking” that it is doing an RBC. This means that the BCR[8] setting will dictate when the WAIT pin deasserts, with this change being referenced to the next clock edge after the access to column 127. Figure 8 shows this behavior.

Figure 8: Single Burst READ Starting at Column 127



Dealing with Losing (First Column of a Row) Data

Some memory controllers, when transitioning across the row boundary, may not be able to handle the relationship between WAIT and the last access (which is for the first column of the next row). One work around is shown here:

1. Start burst WRITE at 0x7Ch (row 0, column 124).
2. Subsequent clock edges control the burst WRITE to locations 0x7Dh thru 0x7Fh.
3. The transition of the WAIT pin will mean that the memory controller writes the value for location 0x80h (row 1, column 0). The memory controller can then do one of two things:
 - Stop the burst by raising CE#, and restart the burst with an address of 0x80h (row 1, column 0). This means that the direct memory access (DMA) resends the 0x80h data.
 - Continue to monitor the WAIT pin, ensure a RBC occurs, and continue the burst WRITE at the next location (row 1, column 1).

Conclusion

When a memory controller attempts burst READ/WRITE operations across an RBC, the main areas to consider are:

- An RBC burst WRITE will be successful if CE# is asserted for (2LC + 1) when the row crossing occurs.
- The WAIT pin indicates when the RBC occurs so the memory controller can take the appropriate action.
- The WAIT pin operation will vary depending on the settings in BCR[8] and BCR[10].

For further technical assistance, e-mail psramsupport@micron.com or visit Micron's Web site: www.micron.com/products/psram.

References

- Micron CellularRAM data sheet—MT45W2MW16B
www.micron.com/products/psram/cellularram

Appendix A

CellularRAM CR1.5 Differences

RBC is not supported by the CellularRAM Workgroup specification for CR1.5 devices and therefore is not outlined as a feature in Micron CR1.5 device data sheets. However, Micron CR1.5 devices do provide RBC functionality as described in this document. Please contact the factory for additional details regarding RBC support on specific Micron devices.

The differences for the WAIT assertion period at RBC for various Micron CellularRAM parts are detailed in Table 5.

Table 5: WAIT Assertion Differences Among Micron Devices

Density	CR	Part Number	Burst Write Delay	Burst Read Delay
16Mb	1.0	MT45W1MW16BA	2LC + 1	2LC
16Mb	1.5	MT45W1MW16BD	LC + 1	LC
32Mb	1.0	MT45W4MW16BA	2LC + 1	2LC
64Mb	1.0	MT45W4MW16B	2LC + 1	2LC
64Mb	1.5	MT45W4MW16BC	LC + 1	LC
128Mb	1.5	MT45W8MW16B	LC + 1	LC

All other RBC entry and exit functionality is the same for WAIT assert/deassert.

Burst Address/Data Multiplexed Considerations

Micron has developed a family of address/data multiplexed (A/D MUX) PSRAM devices. This PSRAM/memory controller interface change does not impact any of the RBC considerations in this technical note. Table 6 lists the duration of RBC WAIT assertions for A/D MUX parts. RBC functionality is defined within the appropriate data sheet for these parts.

Table 6: WAIT Assertion Differences Among Micron Burst A/D MUX Devices

Density	Part Number	Burst Write Delay	Burst Read Delay
16Mb	MT45W1MW16MBP23Z	LC + 1	LC
32Mb	MT45W2MW16MBP24A	2LC + 1	2LC
64Mb	MT45W4MW16MBP25Z	LC + 1	LC



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